

Problem 288: Apportionment

Difficulty: Medium

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Problem Background

Article I, Section 2 of the US Constitution establishes the requirement to conduct a census every 10 years for the purpose of determining each state’s representation in the US House of Representatives. This was last done in 2020, and is expected to be done again in 2030. Each census results in shifts in the congressional balance of power. In 2020, the changes were the smallest seen for a century; California, Illinois, Michigan, New York, Ohio, Pennsylvania, and West Virginia each lost a seat in Congress; those seats were re-assigned to Colorado, Florida, Montana, North Carolina, Oregon, and Texas (which gained two seats).¹

The method the US Census Bureau uses to assign these seats is defined in law (2 U.S.C. §§ 2a-b); it is called the “Equal Proportions Method” and uses a prioritization system to assign seats to states. The US Census Bureau has asked Lockheed Martin to create a software program to assist with implementing this method more quickly.

Problem Description

The US House of Representatives currently contains 435 seats, which are assigned to states based on each state’s population. Each state is guaranteed at least one seat, regardless of its population, meaning that there are **385 seats** that actually need to be assigned. As noted above, this is done according to a priority system, where combination of state and number of extra seats is assigned a priority value. When these priority values get sorted, seats get assigned to whichever state ends up on top of the list, until all of the seats have been assigned.

To start with, you need to calculate a number of multipliers to use for each successive seat. Each multiplier is calculated as follows:

$$M_n = \frac{1}{\sqrt{n(n-1)}}$$

In the equation above, n is the number of seats in the House of Representatives a state would have if assigned another seat; M_n is the multiplier used to determine a state’s priority for being granted that seat. Since all states have one seat by default, the first multiplier we need to calculate is M_2 – the multiplier used to determine if a state is awarded a *second* seat. You’ll need to be sure to calculate

¹ “2020 Census Apportionment Results.” Census.Gov, US Census Bureau, 7 June 2024, www.census.gov/data/tables/2020/dec/2020-apportionment-data.html.

enough multipliers to cover the highest number of seats a state might gain; for reference, the highest number of seats ever held by one state is 53 (California, from 2000 through 2019), but since we'll be giving you made-up data, you will likely need more multipliers than that.

Once you have all of the multipliers calculated, you'll need to multiply each state's population by each multiplier to determine their priority for getting assigned new seats. All of these numbers then get sorted; the highest-ranking states get assigned seats until there aren't any seats left.

Let's use a small example, using just a few states and their populations from the 2020 census:

State	2020 Population	Priority for 2 nd Seat ($M_2 = 0.70710678$)	Priority for 3 rd Seat ($M_3 = 0.40824829$)	Priority for 4 th Seat ($M_4 = 0.288675134$)
Florida	21,570,527	[2 nd] 15,252,665.915	[5 th] 8,806,130.772	[7 th] 6,226,874.785
Pennsylvania	13,011,844	[4 th] 9,200,763.128	[8 th] 5,312,063.069	[9 th] 3,756,195.818
Texas	29,183,290	[1 st] 20,635,702.256	[3 rd] 11,914,028.253	[6 th] 8,424,490.169

With this limited example, Texas has three total seats before Pennsylvania gets a second; Pennsylvania isn't allocated a third seat until both Florida and Texas have each been assigned four. (In fact, if we'd calculated more multipliers, Texas would be given six seats before Pennsylvania got their third.)

As with most of our problems, the official test cases will include randomly generated population data, but you can safely assume that there will be 50 states and 435 seats in the House of Representatives for each of those test cases. For testing this problem, you are encouraged to use historical census data provided from the US Census Bureau at <https://www.census.gov/topics/public-sector/congressional-apportionment.html> to check your solution; since we're using the same approach the Census Bureau uses, if your data matches their data, you've probably got everything correct. Don't use data prior to 1960, though; that's the first census that included Alaska and Hawaii as states, so using earlier data will throw off your counts.

Sample Input

The first line of your program's input, **received from the standard input channel**, will contain a positive integer representing the number of test cases. Each test case will include 50 lines of text, each containing the following values separated by a space:

- The postal abbreviation for a US State, consisting of two uppercase letters. All such values will be actual abbreviations, and abbreviations will not be repeated within a test case. These are not guaranteed to be in any particular order.
- A positive integer representing the population of that state.

Due to the length of the sample input, it is not replicated in full here; please download the sample input from the contest website's "Problemset" tab.

```
1
AL 5030053
AK 736081
AZ 7158923
AR 3013756
[...sample input truncated as noted above...]
WA 7715946
WV 1795045
WI 5897473
WY 577719
```

Sample Output

For each test case, your program must print fifty lines of text, one for each state, indicating the number of seats in the US House of Representatives the state should be assigned based on the given population data. States should be listed in alphabetical order according to the provided postal abbreviation in the following format:

- The provided postal abbreviation
- A space
- A positive integer representing the number of seats to apportion to that state

Due to the length of the sample output, it is not replicated in full here; please download the sample output from the contest website's "Problemset" tab.

```
AK 1
AL 7
AR 4
AZ 9
[...sample output truncated as noted above...]
WA 10
WI 8
WV 2
WY 1
```